



Design Task - Lorry Crane Hydraulics

REPORT FOR DESIGN TASK IN FLUID POWER SYSTEMS

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Academic Year:
2026

Abstract: The following report consists of the design and mathematical process which we underwent to design the hydraulic circuit for a lorry crane. The system contains two cylinder, and it was our task to determine the operating conditions for the cylinders and pump, and pick-out the appropriate components for the task. The dimensioning also required us to consider a work-case scenario, which informed our decision during part-selection.

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1. Design Task Definition

The objective of the design is to design a lorry crane, that can operative with a given mass over a defined drive cycle, and withstand a worst-case loading scenario.

For the task a certain values were given, mainly defining the drive cycle. Some values were true for all group, while others were group specific.

1.1. General Values

The general values for the drive-cycle provided included:

- $v_{p01} \rightarrow$ Normalised velocity vector for piston 1
- $v_{p02} \rightarrow$ Normalised velocity vector for piston 2
- $e_1 \rightarrow$ Level-arm for cylinder 1
- $e_2 \rightarrow$ Level-arm for cylinder 2
- $\theta_1 \rightarrow$ First arm angle
- $\theta_2 \rightarrow$ Second arm angle
- $L_0 \rightarrow$ Base height for lorry crane
- $p_2 \rightarrow$ Pressure on low-pressure side
- $\eta \rightarrow$ Oil dynamic viscosity
- $c_v \rightarrow$ Laminar leakage loss coefficient
- $k_p \rightarrow$ Coloumb friction coefficient
- $k_v \rightarrow$ Viscous friction loss coefficient
- $k_\epsilon \rightarrow$ Displacement coefficient

The information on the work-case scenario, provided the following values:

Parameter	Cylinder 1	Cylinder 2
$\theta_{1,wc}[\text{rad}]$	1,2	0,35
$\theta_{2,wc}[\text{rad}]$	0,0	0,0
$e_{1,wc}[\text{m}]$	0,14	0,48
$e_{2,wc}[\text{m}]$	0.57	0,17

1.2. Group-specific values

There were many group-specific values that we were given for the design task, these included:

- $M_{t,wc} \rightarrow$ Maximum total load mass
- $q_{r,wc} \rightarrow$ Worst-case flow to drive-cycle-max flow ratio
- $M_t \rightarrow$ Total load mass
- $L_1 \rightarrow$ First arm length
- $L_2 \rightarrow$ Second arm length
- $v_{p0} \rightarrow$ Reference cylinder velocity
- $t_0 \rightarrow$ Drive-cycle Time
- $P_{s,max} \rightarrow$ Maximum system pressure
- $\eta_c \rightarrow$ Cylinder Efficiency

1.3. Denormalisation and discretization

From the provided values we can see that the velocity profile which we were provided has been normalised and we have been provided a reference value. Using this reference value we denormalised the normalised velocity profile, to determine the effective velocity profile of the two cylinders.

$$v_{p1} = v_{p0} * v_{p01} v_{p2} = v_{p1} * v_{p02}$$

Additionally, we discretized time to a number of discrete times equal to the length of the provided vectors for velocity, angles and arms, as such:

$$\text{time} = \text{linspace}(0, t_0, \text{length}(e_1));$$

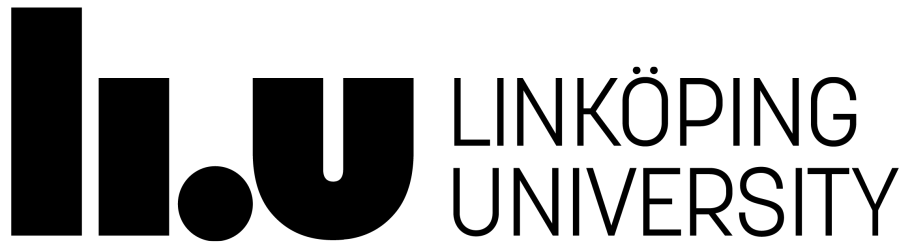
2. Cylinder Sizing and Selection

blablabla

3. Circuit Drive-cycle Analysis

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$$q = C_q A \sqrt{\frac{2}{\rho} \cdot \Delta p} \quad (1)$$



4. Pump Selection

5. System Analysis and Efficiency

6. Cooler Dimensioning